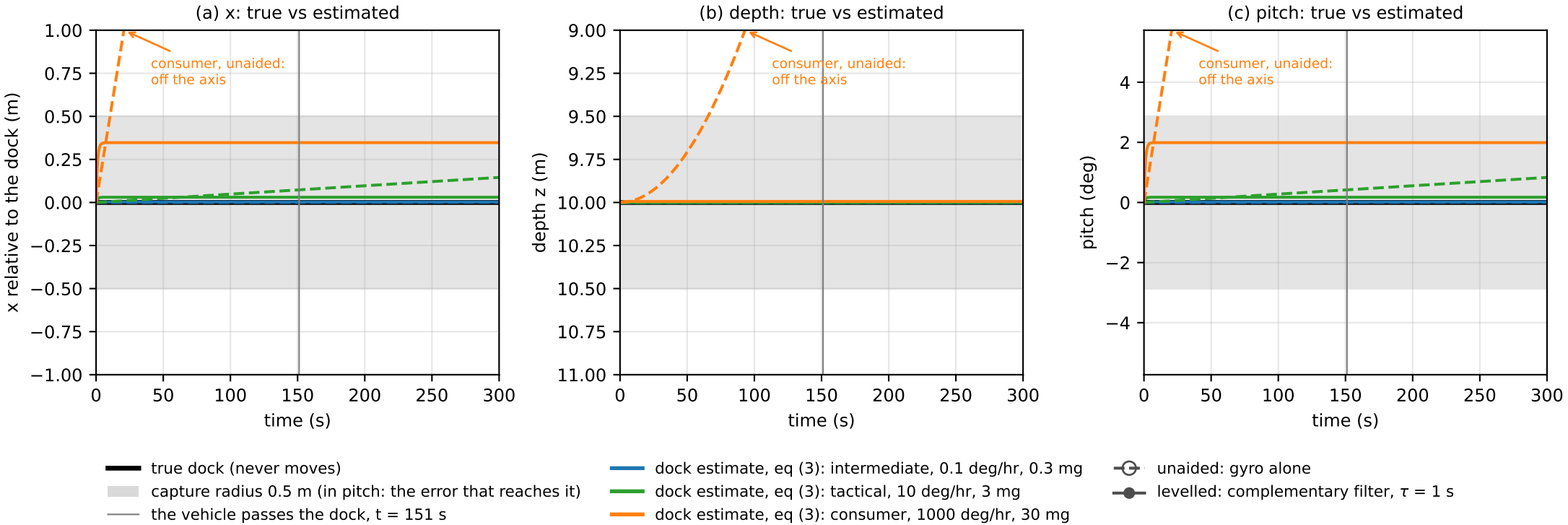
Dock position estimated through the cable, eq (3), from the attitude its own navigation unit gives,
for three IMU grades - black is the truth, which never moves
(a) x: true vs estimated
(b) depth: true vs estimated
(c) pitch: true vs estimated
consumer, unaided:
off the axis
x relative to the dock (m)
depth z (m)
pitch (deg)
time (s)
true dock (never moves)
capture radius 0.5 m (in pitch: the error that reaches it)
the vehicle passes the dock, t = 151 s
dock estimate, eq (3): intermediate, 0.1 deg/hr, 0.3 mg
dock estimate, eq (3): tactical, 10 deg/hr, 3 mg
dock estimate, eq (3): consumer, 1000 deg/hr, 30 mg
unaided: gyro alone
levelled: complementary filter, τ = 1 s
IMU grades from Groves' table of typical biases (consumer: one decade past its lower bounds). The dock's INS starts from its true attitude (alignment assumed perfect).
Its gyro bias is about the pitch axis and its accelerometer bias along x, so y, roll and yaw stay exact. The vehicle and buoy have ideal IMUs.